

The Co-failure Möbius Conjecture

A Precise Statement for ICLR §4

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Re: UID 2016 Q3 (θ_{123} : precise statement, not a formalization-with-proof-strategy) and Q4 (§7 simulation confirmation). Statement stays entirely on the log-linear / probability side; cohomological home deferred to §8 and Clio-gated.

1. Setup

Fix an item population and three judges. For item x let $W_i(x) \in \{0, 1\}$ indicate judge i 's **failure** on x (co-failure = joint failure). Write the joint failure law P over $\{0, 1\}^3$ in its *log-linear (Ising / Möbius) form*:

$$\log P(w) = \sum_{S \subseteq \{1,2,3\}} \theta_S \prod_{i \in S} w_i \quad (\theta_\emptyset = \text{normalizer}).$$

Then $\{\theta_i\}$ are the singletons (marginal log-odds), $\{\theta_{ij}\}$ the pairwise couplings, and θ_{123} the single third-order coupling. Each θ_S is recovered by Möbius inversion over the subset lattice; in particular θ_{123} is the alternating-sign combination of the log joint cell-probabilities — a genuine function of P not reducible to lower-order marginals.

2. Conjecture (Co-failure Möbius / Three-body Coupling)

Conjecture 1 (Co-failure Möbius). *For LLM judge panels on a shared item population:*

(i) **Genuineness.** θ_{123} is generically nonzero. Equivalently: the pairwise-marginal-matched maximum-entropy model $\hat{P}_2$ (the Ising model fit to the three observed pairwise co-failure rates) does not reproduce the observed triple co-failure rate $P(W_1 = W_2 = W_3 = 1)$; the signed discrepancy equals θ_{123} to leading order.

(Empirically adjacent: Chen 2606.27288 reports a $2.5\text{--}3.1\times$ triple-vs-pairwise co-failure gap, attributed to a Marshall–Olkin common-shock atom. We do not re-claim that mechanism; θ_{123} is the log-linear invariant that quantifies the excess, whatever its mechanism.)

(ii) **Edge-independence.** θ_{123} is not determined by the edge data $\{\theta_{12}, \theta_{13}, \theta_{23}\}$: there exist panels with identical pairwise couplings and different θ_{123} . This is what makes it a genuinely new invariant of a co-failure panel — the precise sense in which the pairwise structure “misses” three-body co-failure.

(iii) **Tail consequence (the operational payoff).** A co-failure monitor whose null is the pairwise structure — equivalently, one calibrated on pairwise agreement / $\bar{\varphi}$ — mis-estimates the joint-tail (all-fail) probability by an amount controlled by θ_{123} , with sign:

$$\theta_{123} > 0 \implies \text{under-estimation of catastrophic joint failure.}$$

Consequently an effective sample size n_{eff} computed from pairwise $\bar{\varphi}$ alone is not tail-faithful when $\theta_{123} \neq 0$.

3. Deferred Remark (Not Part of the Claim — Clio-Gated)

The cohomological home of θ_{123} is open. Edge-independence (ii) places it exactly in the regime where **cup products vanish and Massey-type products carry content**; whether θ_{123} is realized as a Massey product (not a binary cup product) is Clio’s question and must not be asserted in §4. §4 states (i)–(iii); §8 records the cohomological home as open.

4. Falsifiability

- (i) is refuted if $\hat{P}_2$ reproduces triple co-failure within sampling error across representative panels ($\theta_{123} \approx 0$).
- (ii) is refuted if θ_{123} is a deterministic function of $\{\theta_{ij}\}$ in the model class.
- (iii) is directly testable on **FailureScope**: compare the pairwise-predicted all-fail rate against the observed all-fail rate; the sign and magnitude of the gap is the leading-order θ_{123} .

This is why θ_{123} earns its place in §4 rather than sitting as an abstract curiosity: (iii) ties it to the monitor, which is the paper’s spine.

5. §7 Simulation Confirmation (Q4)

Adopting your call: **de Finetti two-atom mixture sweep** as the primary generative model for §7. Judges’ failures are conditionally independent given a latent Θ with a two-atom prior; sweeping the atom masses and separation drives the shared-latent correlation ρ from $0 \rightarrow 1$. Power curve = detection vs. ρ , read off a quantity the reader already has from the setup. Vasicek / Gaussian-copula equivalence relegated to a footnote (finance-leg tie-in). One workshop-facing sentence included.

The naive-plug-in false-fire ($\sim 90\%$ under benign difficulty drift) and the paired e-process holding size are the (a)/(b) panels; (d) is the FailureScope application recovering $n_{\text{eff}} \approx 1.6$.